

Structure and Response of Spherical Diffusion Flames

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Outline

- Summary Introduction
- Experimental Objectives
- Experimental Approach to Achieve Objectives
- Justification for Extended Microgravity Duration
- Other Requirements – Burner
- Operation Sequence (Experiment Plan)
- Summary of Test Matrix Parameters
- Success Criteria



Goals

- Use spherical diffusion flames in long-duration μg environment to:
  - obtain experimental data of exceptional quality in simple, well-defined flow fields.
  - understand various key elemental diffusion flame processes.
- Compare these data with those obtained through independent computational simulation with detailed chemistry and transport in order to:
  - scrutinize and improve the accuracy and comprehensiveness of these descriptions.



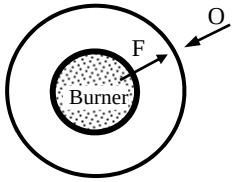
Merits of One-Dimensional Flames for Fundamental Studies

- Flame structure fundamentally affected by aerodynamics.
- Simple flow and flame configuration facilitate experimentation, computation, analysis, and comparison, with enhanced accuracy and reliability.
- Reduced effort in describing flows allows more detailed study of other aspects of flames such as chemistry.
- A one-dimensional flame is the simplest possible configuration.

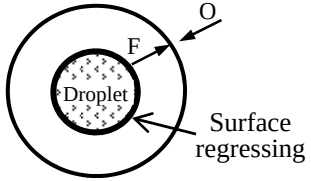


Merits of Burner-Generated Spherical Diffusion Flames

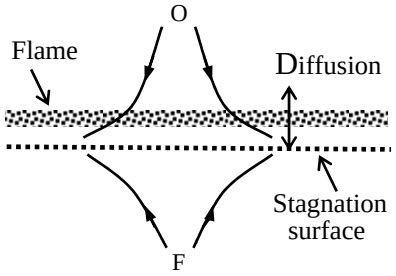
- Only spherical (vs. planar and cylindrical) configuration admits steady-state 1D behavior in semi-infinite domain.
- Spherically symmetric droplet combustion complicated by transient effects due to droplet heating and surface regression. Kinetics of large HC fuels also less well established.
- Burner generation assures steady-state surface boundary conditions.



Spherical Burner Flame



Droplet Flame



Counterflow Flame

- Planar flame in counterflow is not 1D in terms of diffusion-convection as well as radiation misalignment.



Focus A: Experimentation on Elemental (Diffusion) Flame Processes

- Spreading of diffusion flame sheets
- Dual extinction states
- Flamefront instabilities

Focus B: Development of Fundamental Physical & Chemical Database

- Detailed fuel oxidation mechanisms
- Reduced fuel oxidation mechanisms
- Diffusion coefficients
- Radiation models



The Role of μg Spherical Diffusion Flame in Comprehensive Mechanism Development

- Substantial research on homogeneous systems (shock tubes, flow reactors, *etc*.) and premixed flames
- Diffusion flame experiments mostly involve counter- and stagnation flows, and droplet flames
- Burner-supported spherical flame is 1D and steady state
- Experiments of interest to chemistry are those involving ignition and extinction.
- Experiments for situations away from extinction would yield information on the diffusion and radiation parameters.



Article Published Since SCR

Yang, F., Law, C.K., Sung, C.J., Zhang, H.Q. "A mechanistic study of Soret diffusion in hydrogen-air flames" *Combustion and Flame* **157**, 192-200, 2010.



Experimental Objectives

Objective A. **Flame Structure Evolution:** Measure and characterize the transient structure (such as temperature and species distribution) of diffusion flames established at initially non-steady, spherical flame structures for mixtures of different H_2/CH_4/diluent ratios (soot-free) and different C_2H_4/diluent ratios (sooty), for different flow rates.

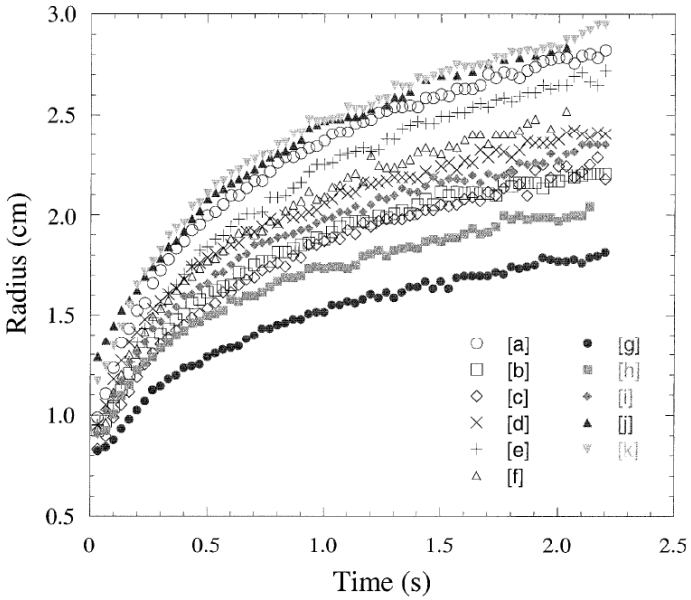
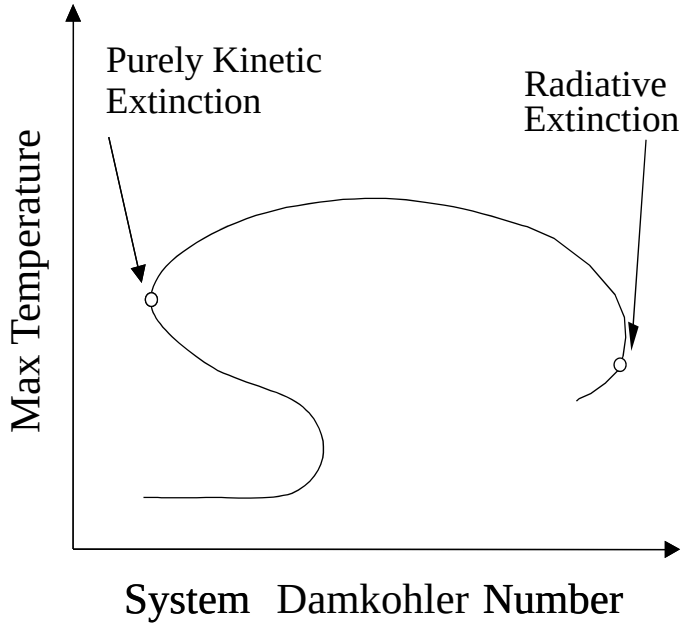


Fig. 3. Data of flame radii as a function of time for various cases. *20%H_2-25%CH_4-55%N_2*: [a] 15.72 mg/s. *25%H_2-20%CH_4-55%N_2*: [b] 10.55 mg/s. *30%H_2-15%CH_4-55%N_2*: [c] 9.42 mg/s, [d] 12.0 mg/s, [e] 15.31 mg/s. *40%H_2-10%CH_4-50%N_2*: [f] 12.37 mg/s. *50%H_2-10%CH_4-40%N_2*: [g] 4.96 mg/s, [h] 6.43 mg/s, [i] 8.1 mg/s, [j] 1.67 mg/s, [k] 13.03 mg/s.



Experimental Objectives

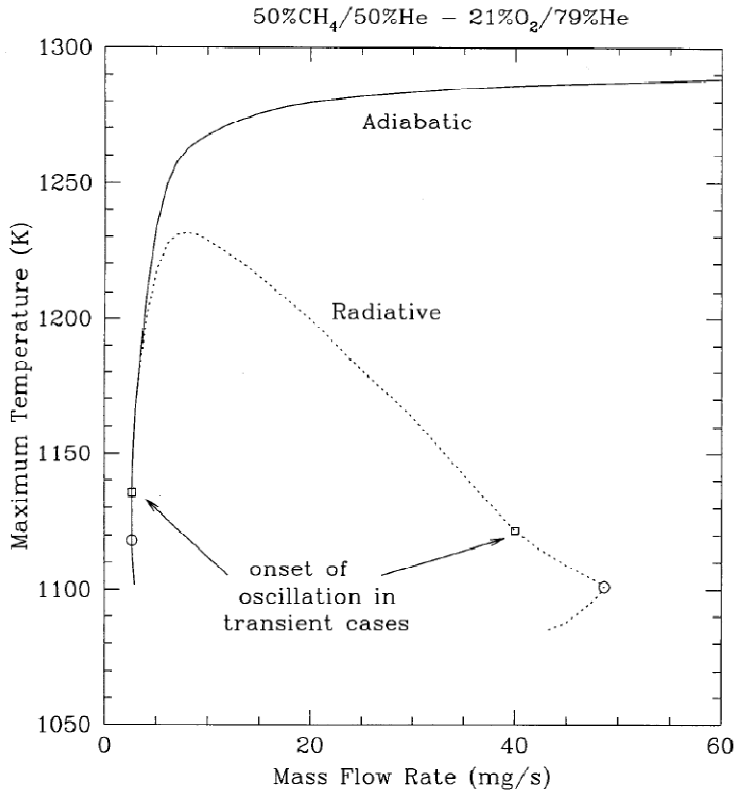
Objective B. Dual Extinction Limits at Low and High System Damköhler numbers, Da: Determine the quasi-steady convective/chemical-kinetic extinction limits (low system Da) and radiative/chemical-kinetic extinction limits (high system Da) for mixtures of different H_2/CH_4/diluent ratios (soot-free) and different C_2H_4/diluent ratios (sooty).





Experimental Objectives

Objective C. **Flame Front Instabilities in Non-Premixed Flames:** Determine the existence (for different Zel'dovich number, Ze), onset (as a function Da), and nature (i.e. mode, frequency, amplitude) of pulsating instabilities theoretically predicted to occur in transient spherical diffusion flames using fuel/diluent mixtures that are above a critical Lewis number.

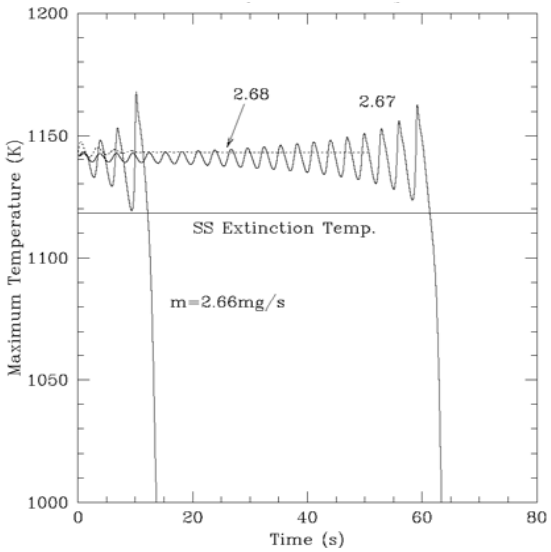




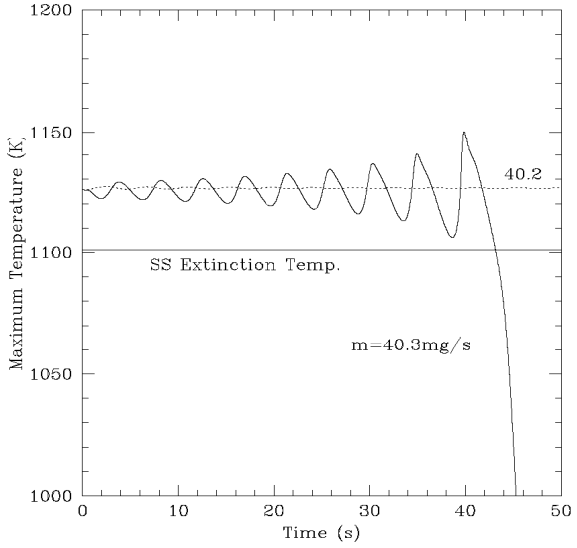
Oscillatory Response Prior to Extinction

- 50%CH_4 / 50%He vs. air (Le_F = 2.3, Le_o = 1.9)
- Oscillation observed at both kinetic and radiation extinction limits

Kinetic Extinction



Radiation Extinction

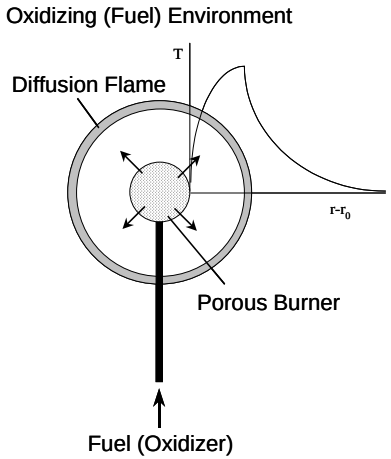


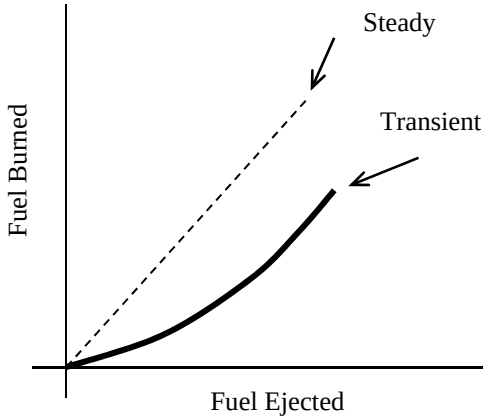
Computation



Experiment Setup and Approach

- A series of experiments is proposed in which mixtures of various gaseous fuels, diluents and oxygen will be issued into an oxidizing environment through a spherical porous burner and ignited to nominally obtain a spherical, one-dimensional stabilized flame
- All experimental sets will address transient phenomenon, as flame evolves from localized ignition towards “steady” behavior





Flame spread behavior

- Various mixtures to assess both chemical and transport effects on flame behavior



Achieving Objective A

- **Objective A – Flame Structure Evolution**
  - Fuel/diluent mixtures will be issued from the spherical burner and ignited at flow rates expected to form flames that evolve toward idealized steady-state conditions.
  - As the flame spreads outward, transient measurements of flame front velocity and flame structure will be obtained using imaging and radiometric systems.
  - The flame characteristics will be compared with the predictions of the transient numerical simulation of spherical diffusion flames for evaluations of detailed chemical kinetic and transport models.

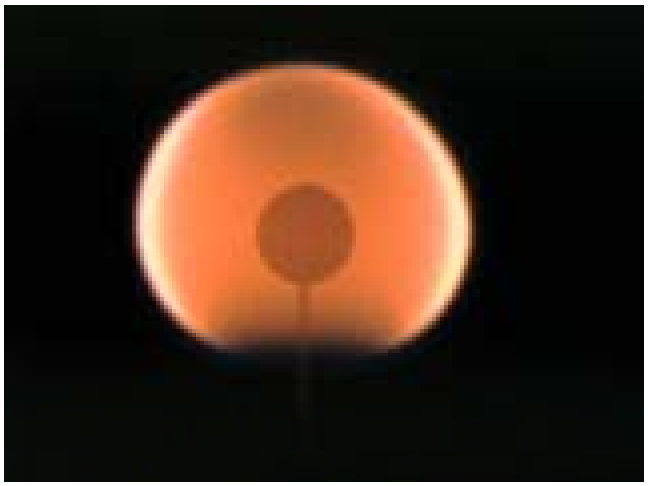


Flame Structure Evolution SDEP A1 - Direct Color (Visible) Imaging

- Sequence of color and OH* images of the flame as a function of time, flow rate, and fuel composition.

- Visible Chemiluminescence (CH*)
- Soot incandescence
- Assess flame symmetry and influence of feed tube

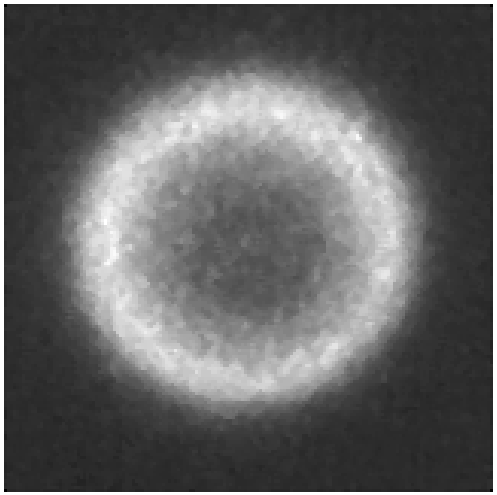




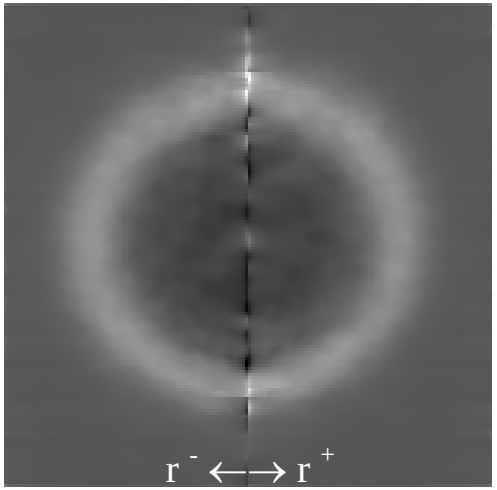


Flame Structure Evolution SDEP A1 - UV Imaging

- Chemiluminescence from OH* (310nm)
  - Chemistry (comparisons with computations)
  - Indirect assessment of flame structure



Projection Image (310nm)



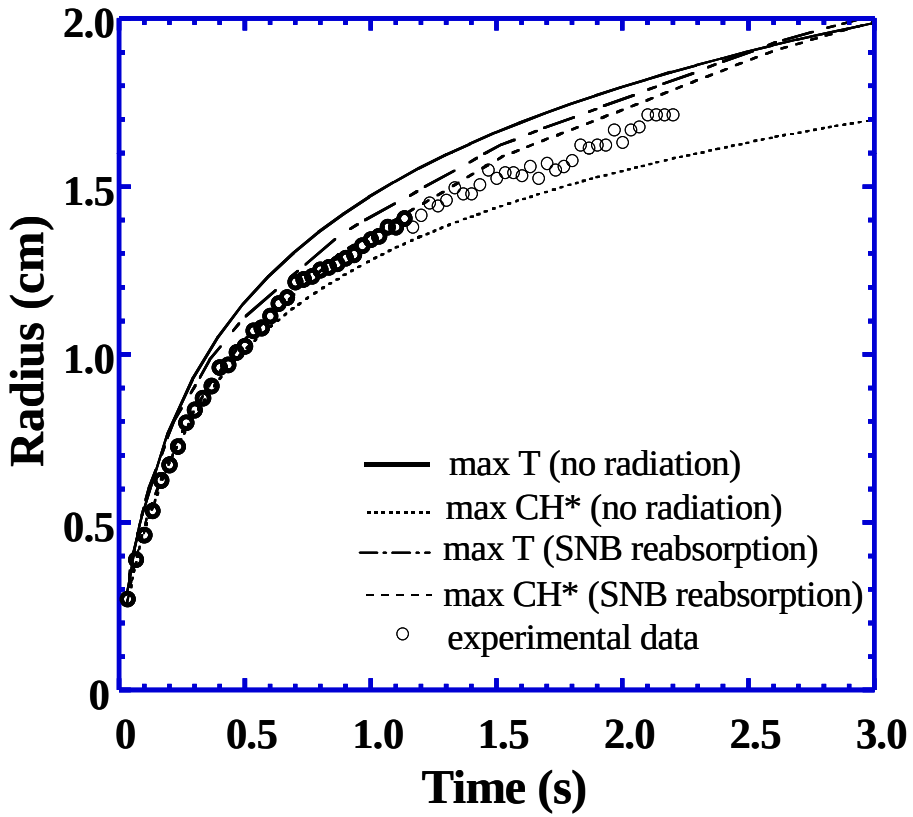
Abel deconvoluted image

- 9%O_2/91%N2, m = 0.078 g/s, into 100%H_2 at 0.079 atm
- Xybion IMC-201 intensified multispectral video camera with filter
- OH* chemiluminescence



Flame Structure Evolution SDEP A2

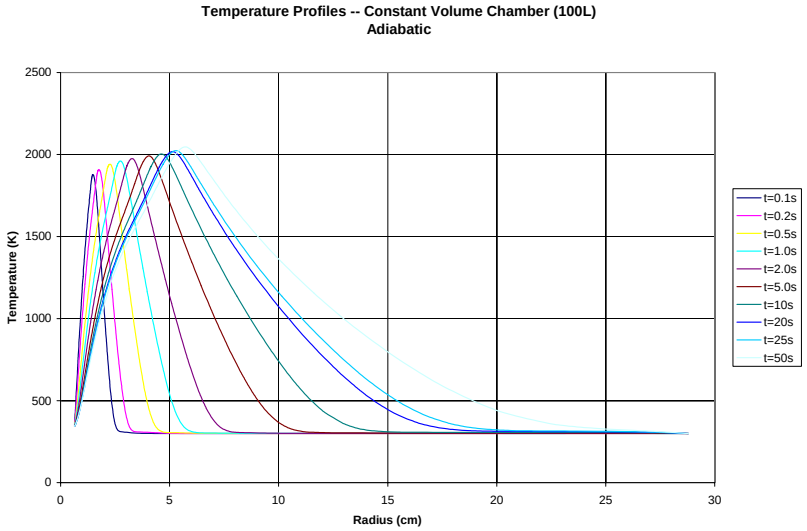
- Visible flame radius as a function of time, flow rate, and fuel composition.
- Requirements
  - Color images
  - UV images

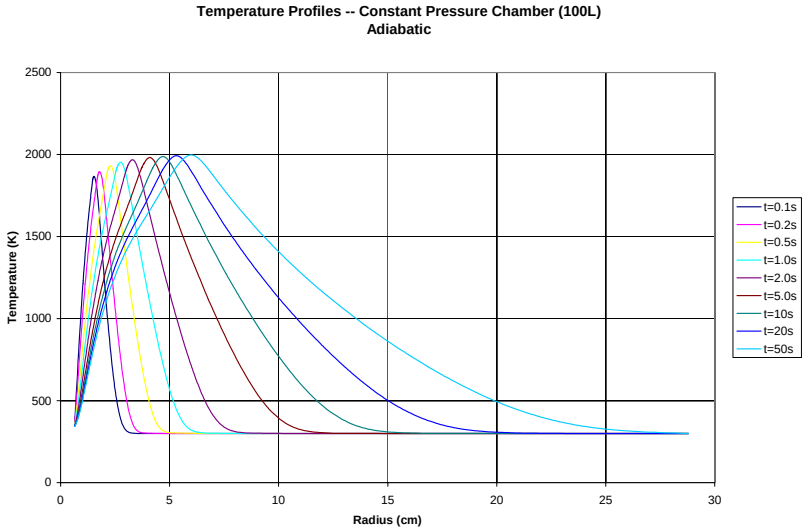




Flame Structure Evolution SDEP A3

- Temperature (and desired major species) radial distribution as a function of time, flow rate, and fuel composition.
- Requirements
  - Temperature profiles





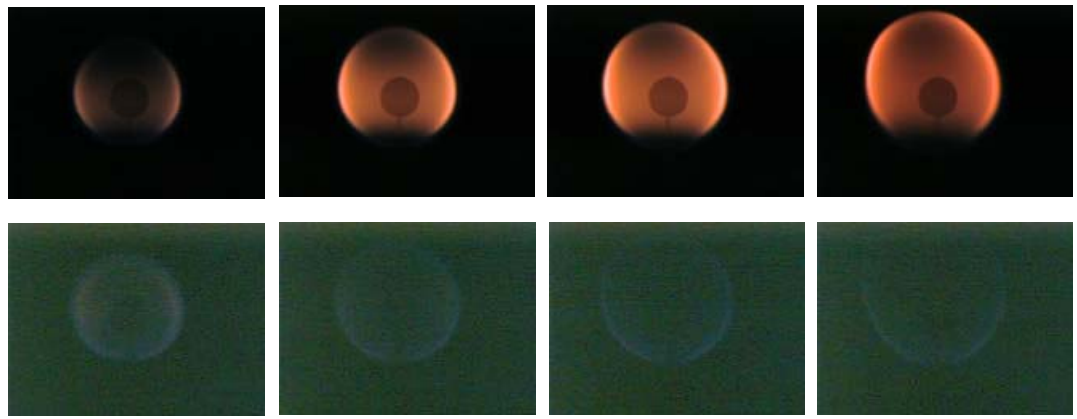
20%H_2-25%CH_4-55%N_2 into air, m=15.72mg/s



Flame Structure Evolution SDEP A4

- Soot onset (determined from visible luminescence versus soot extinction measurements) as a function of radii for different flow rates and fuel compositions.
- Requirements
  - Color images
  - Soot volume fraction measurements
  - Temperature profiles

e.g. 25%C_3H_8 / 2%O_2 / 49% Ar / 24% N_2



High flow rate

Low flow rate



Achieving Objective B

- **Objective B – Quasi-Steady Extinction Limits, Low Da**
  - Fuel/inert mixtures will be issued from the spherical burner and ignited at initial flow rates expected (or known/observed) to form flames that evolve toward idealized steady-state conditions.
  - Following ignition and initial flame front movement, the fuel/inert flow rate will be abruptly reduced to a second flow rate below the expected extinction limit.
  - Quasi-steady extinction is expected to occur as the flame adjusts to the reduced flow rate.
  - To refine the determination of the extinction flow rate for each fuel/diluent mixture, repeated tests are to be conducted at successively lower initial flow rates and higher secondary flow rates.



Achieving Objective B (cont.)

- **Objective B – Quasi-Steady Extinction Limits, Low Da**
  - At the low Da extinction limit, chemical kinetic rates are retarded with respect to flow rates by inert gas in the fuel stream that reduces the flame temperature.
  - Based on theoretical predictions, fuel/inert mixtures are to be chosen for which the extinction flame radius is large enough to minimize the effects of heat loss to the burner.
  - The surface temperature of the burner will be directly measured to quantify this loss.



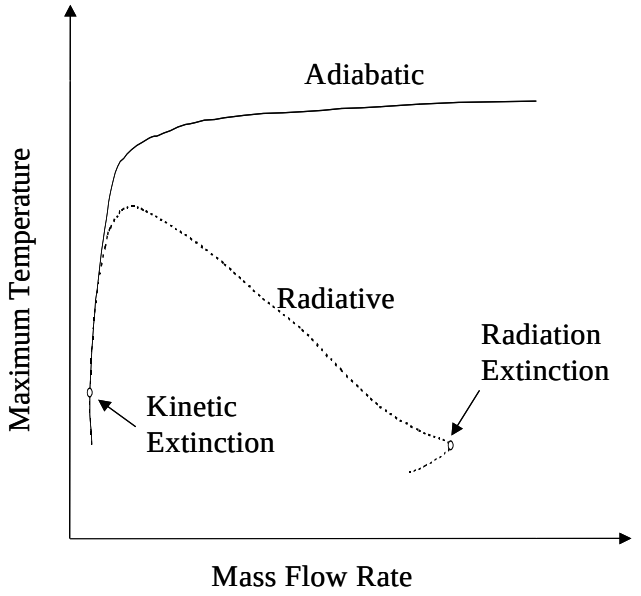
Achieving Objective B (cont.)

- **Objective B – Quasi-Steady Extinction Limits, High Da**
  - Fuel/inert mixtures will be issued from the spherical burner and ignited at flow rates *above and beyond* those expected (or known/observed) to form steady flames when radiative loss is considered.
  - The quasi-steady extinction is expected to occur as the flame moves outward before reaching the *adiabatic* steady-state radius.
  - Rates of radiative emissions from the flame and radiative reabsorption elsewhere are to be accentuated in some cases by using a radiatively active diluent such as CO_2.



Extinction Limits SDEP B1, B2, B4

- B1) Extinction limits as a function of system Damköhler number.
- B2) Total radiation loss as a function of radius for different flow rates and fuel compositions.
- B4) Radiation loss versus time for sooty flames, for different flow rates and fuel compositions.
- Requirements
  - Temperature profiles
  - Color images
  - UV images
  - Mass flow rate
  - Radiometer data



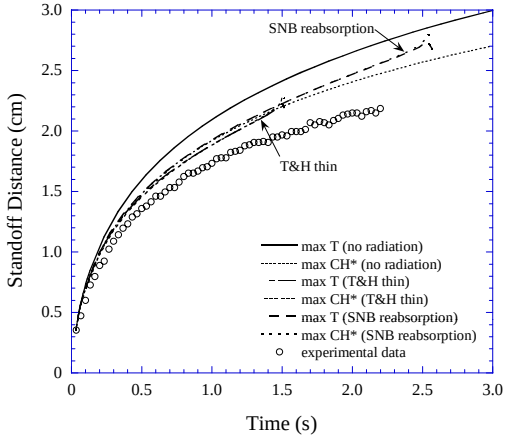


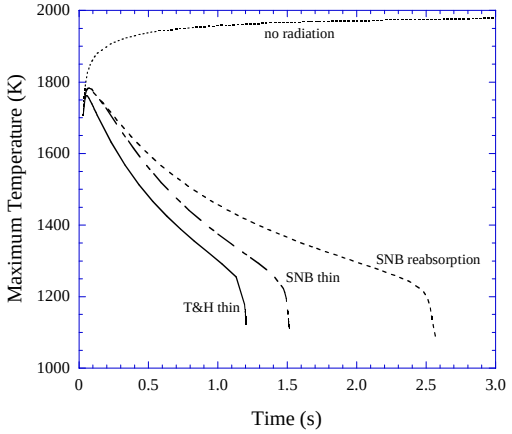
Extinction Limits SDEP B3

- Peak CH* or OH* chemiluminescence along with peak temperature as a function of radius, for different times, flow rates, and fuel compositions.
- Requirements
  - UV images (OH* filter)
  - Temperature profiles
    - Assess evolving flame structure
    - Assess moment of LOCAL flame extinction (based on precipitous drop in local peak temperature)
    - Track flamefront motion based on peak temperature
    - Radial temperature profile to be provided by thin filament pyrometry

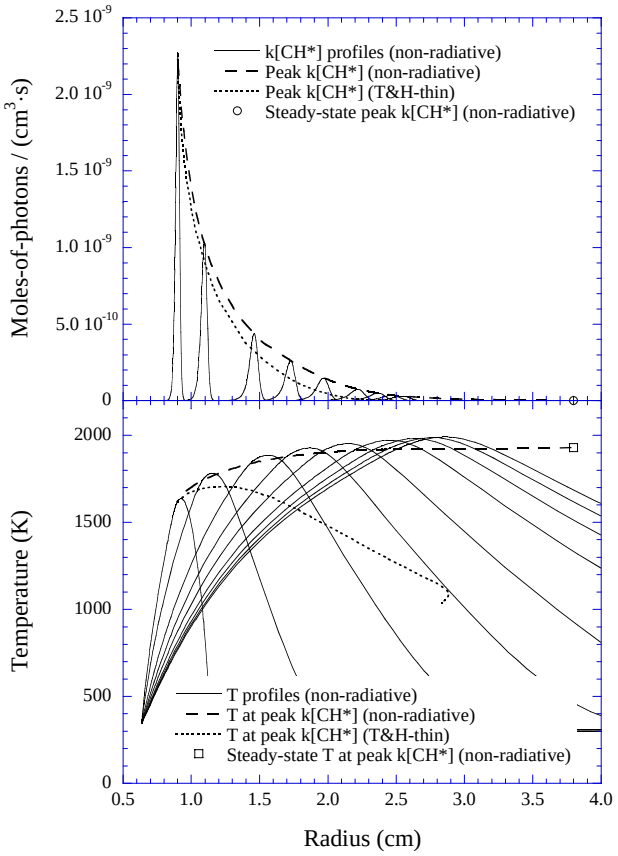


Extinction – Quasi-steady





Visible Luminosity Consideration



- $20\%H_2$-$25\%CH_4$-$55\%N_2$ (15.72mg/s) into STP air



Achieving Objective C

- **Objective C – Flame Front Instabilities**
  - Pulsating flame front instabilities are predicted to occur near the extinction conditions at both low and high Da.
  - The approach taken to observe the instabilities is similar to the approach for observing extinction except fuel/diluent mixtures with large system Lewis numbers (Le) are used.
    - Low Da: moderate initial flow rate, abrupt change to flow rate below predicted extinction rate.
    - High Da: initial flow rate above predicted extinction flow rate with radiation.
    - Instabilities are expected to appear before extinction.
  - Radiating diluents are used to further encourage the instabilities.



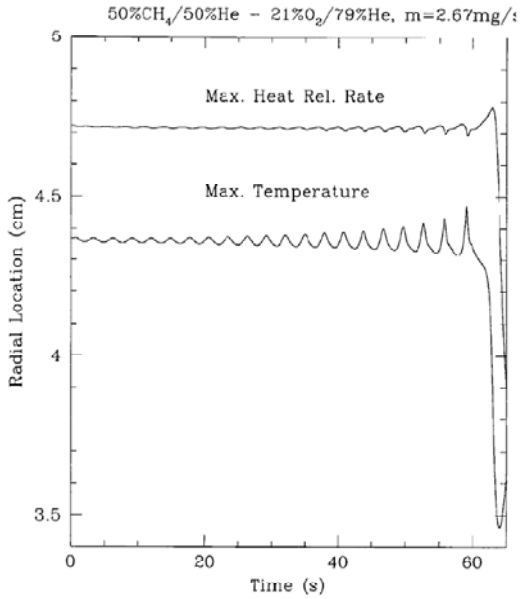
Instabilities SDEP C1, C2, C3, C4

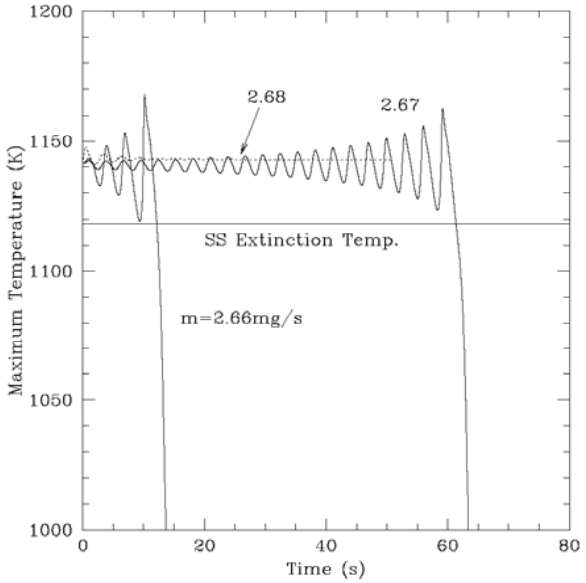
- C1) Total chemiluminescence as a function of time.
- C2) Total radiation loss (based on radiation flux) as a function of radius, for different flow rates and fuel compositions.
- C3) Oscillation mode, frequency, and amplitude near extinction as a function of flow rate for flames of different Le and mass flow rates.
- C4) Comparison of extinction times and extinction radii as a function of flow rate, for different fuel compositions.
- Requirements
  - PMT (chemiluminescence)
  - Radiometer data
  - Color images
  - UV images
  - Temperature profiles
  - Mass flow rate



Radiation Flux Measurement

- Global/local flame radiation measured at faster rate than cameras to capture “high” frequency oscillation





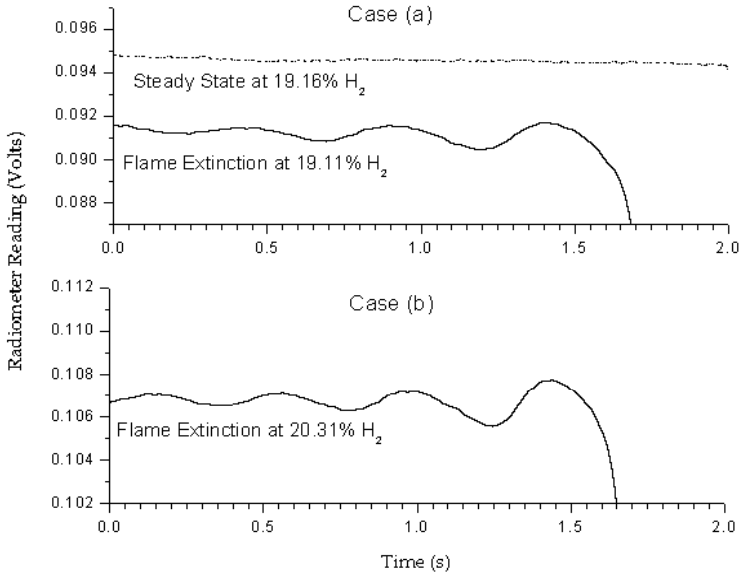
- Assess radiation heat transfer
- Flame movement often small
- Help to assess extinction moment



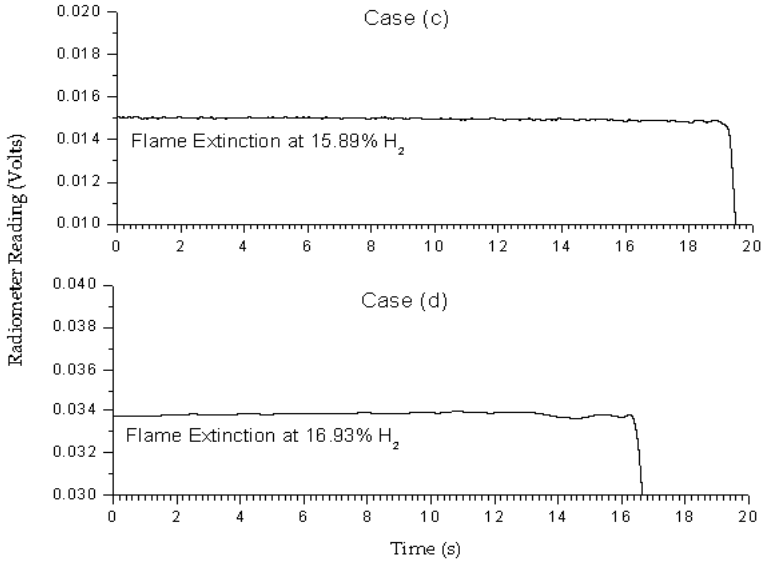
Extinction - Oscillatory

Micro-buoyancy Experimentation (Fuel: Hydrogen)

Radiation Extinction



Kinetic Extinction





Instabilities SDEP C5

- Soot radial distribution as a function of time for different flow rates and fuel compositions during fully transient flame front movement
- Requirements
  - Color images
  - Temperature profiles
  - Soot volume fraction measurements



Justification for Space-Flight Facility

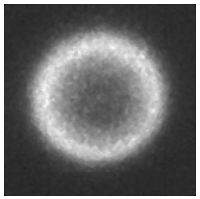
- High-quality and sufficient duration microgravity
  - Transient evolution to “steady-state”
  - Radiative extinction
  - Flame instability
  - Multi-staged flow for kinetic extinction
- Computation comparison (need 1-D flames)
- Diagnostics afforded, KEY → Flame structure
  - UV imaging
  - Radiometer
  - Temperature profile



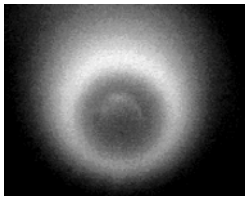
Limitations of Ground-Based Testing

- Inverse-flame configuration in n-*g* is limited and constrained

$9\%O_2/91\%N_2$, m = 0.078 g/s, into $100\%H_2$ at 0.079 atm

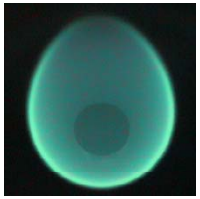


OH* chemiluminesence

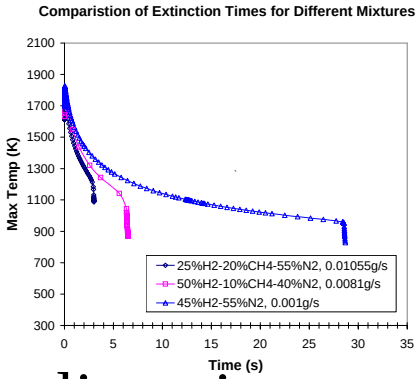


H_2O infrared emission

Air into CH_4 at 0.079 atm



- Durations in 2.2-s and 5.0-s drop towers insufficient to observe radiative extinction
- Radiative extinction never observed in parabolic aircraft (up to 10s low-*g*) → *g*-jitter, $Gr \sim l^3$



- Need extended duration and high-quality microgravity to study diffusion flame oscillations with "low" characteristic frequency



Other Requirements - Burner

- PI provided spherical porous bronze burner can presently provide 0.99 sphericity and 0.94 concentricity diffusion flames.



50%H_2/10%CH_4/40%N_2, 8.1mg/s into STP air (2.2-s drop tower), ½" burner diameter

- Burner purpose is only to provide uniform, spherical flow

Independent of burner size and porosity ($r_f > R_b$)

$$r_f = \frac{\Lambda_O}{\ln\left(\frac{fY_{O\infty}}{Y_{FR}} + 1\right)}$$

- ¼" (0.635cm) diameter burner preferred to minimize burner effect (i.e. achieve small flame dimensions, reduce heat loss to burner)



Operation Sequence

I. Establish chamber at constant pressure and species concentration

II. Discharge fuel mixture through porous burner at initial flow rate

III. Ignite fuel mixture and retract upon successful ignition

IV. Allow for spherical diffusion flame to evolve

V. Apply one of the following:

- Maintain flow rate with same initial mixture for duration of experiment (25s) or until extinction
- Single step decrease in flow rate with same initial mixture at specified time, maintaining new flow rate for duration of experiment (25s) or until extinction

VI. Terminate fuel supply to burner after 25s.



Experiment Plan

Prioritization and Logic

a) Transient phenomenon leading toward steady flame

b) Extinction phenomenon at “quasi-steady”-states

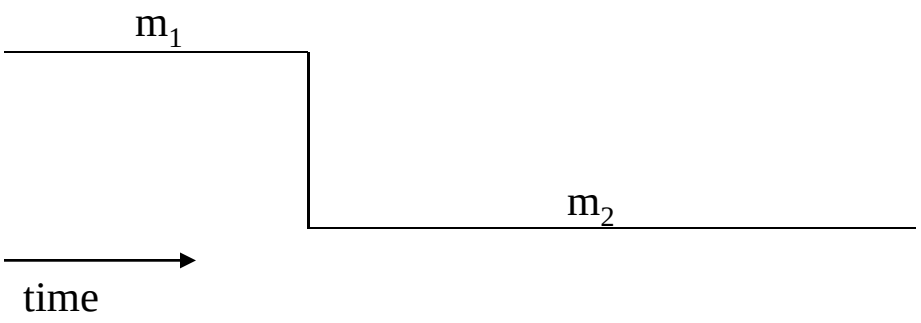
c) Instability phenomenon, e.g. Le > 1 oscillations

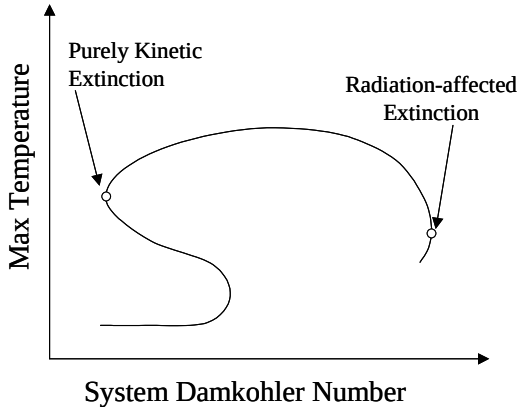
→ Non-sooty and sooty flames



Flow Supply and Flow Rate

- Deliver fuel and inert(s) to burner
- Different flow rates
  - Small versus large characteristic flame sizes (residence time)
    - Radiative extinction
    - Characteristic time to steady state
  - Ignite at moderate flow rate, establish quasi-steady flame, step flow rate down to examine kinetic extinction at small flame radii





- Provide gas (different compositions) at max flow rate for > 25s



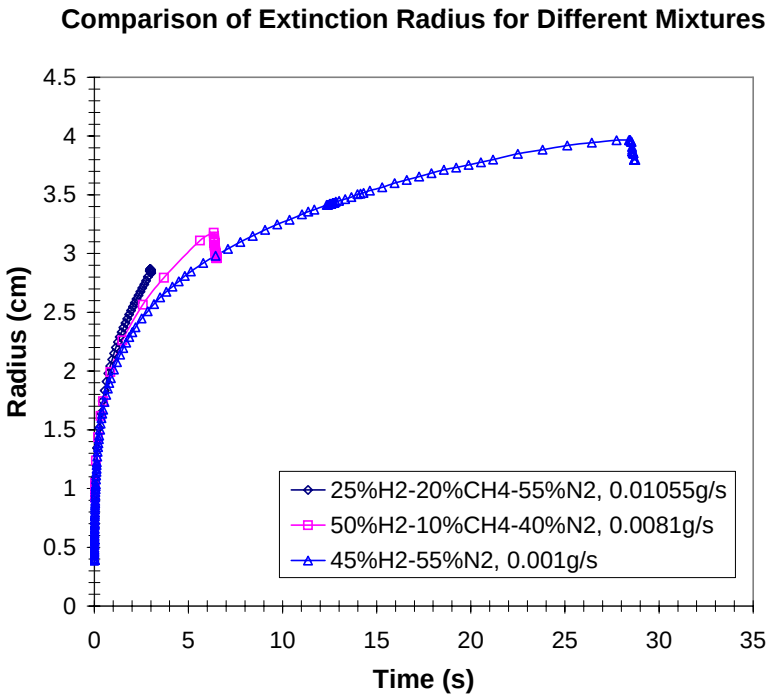
Test Matrix - Fuel Mixtures

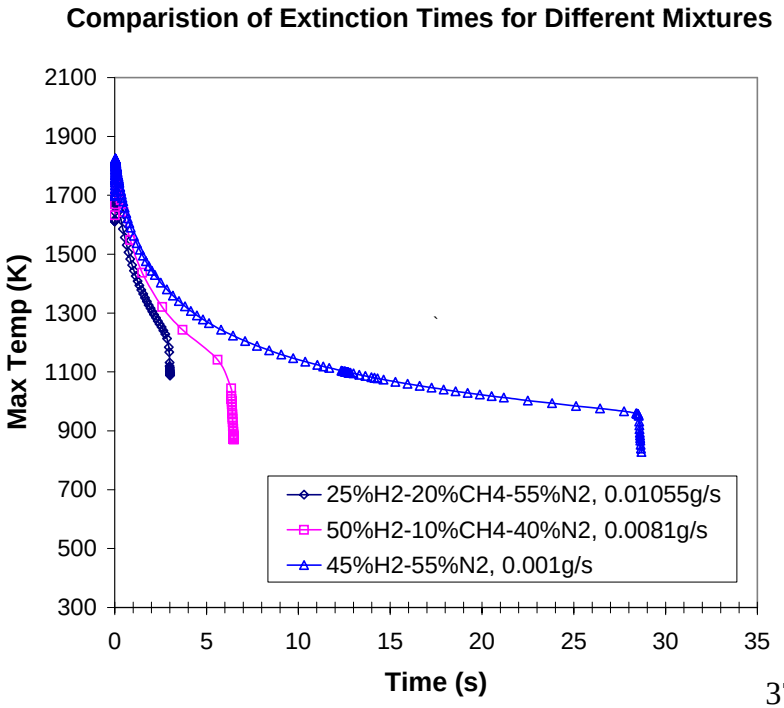
- H_2/Inert (**Required**)
  - Addresses pure hydrogen chemistry
  - Only H_2O in radiation considerations
  - Only OH* chemiluminescence
- CH_4/Inert (**Required**)
  - Addresses methane chemistry, e.g. GRI-Mech
  - H_2O, CO_2, CO in radiation considerations
- H_2/CH_4/Inert (**Required**)
  - Addresses chemical kinetics of blended mixture including kinetic extinction
  - Different characteristic radiative extinction times
  - OH* and CH* chemiluminescence
- C_2H_4 (**Highly Desired**)
  - Nominally sooty C2 fuel
  - Sooting tendencies depend on characteristic residence time, so different flow rates must be examined



Fuel Mixtures

- 20%H_2/25%CH_4/Inert, 25%H_2/20%CH_4/Inert, 30%H_2/15%CH_4/Inert
  - Addresses chemical kinetics of blended mixture including kinetic extinction
  - Different characteristic radiative extinction times







Test Matrix - Inerts

- Different inerts to affect diffusive properties, radiative properties, and flame temperature
  - N_2 (**Required**)
    - Moderate flame temperature
    - Radiatively transparent
    - Normal “air”
  - He (**Required**)
    - Higher flame temperature
    - Radiatively transparent
    - High-Lewis number promoting pulsating instability
  - CO_2 (**Desired**)
    - Lower flame temperature
    - Optically participating medium (highly reabsorptive species)
- All inerts fixed at 0.55 mole fraction for non C_2H_2 fuel mixtures



Test- Matrix Ambient Environment

- STP
- Different sets of ambient gas mixtures to alter characteristic flame temperatures, diffusive transport, and radiative properties
  - "Air": 0.21 O_2, 0.79 N_2 (**Required)**
  - "He-Air": 0.21 O_2, 0.79 He (**Required**)
  - "CO_2-Air": 0.21 O_2, 0.79 CO_2 (**Desired**)
- Specified to within 1% mole fraction
- Ambient gas may be scrubbed and recycled for subsequent tests



Minimal Success

- Observations of transient flame phenomena leading toward steady state flames for 1 CH_4 case and 1 C_2H_4 case; or
- Observations of flame extinction at both low and high system Damköhler number for 1 case; or
- Observations of spherical soot formation in 1 C_2H_4 case.



Significant Success

- Observations of at least two instances of flame front instabilities including at least one hydrocarbon fuel; or

- Combinations or extensions (in terms of number of mixture ratios for a single hydrocarbon fuel or number of hydrocarbon fuels) of at least two of items



Complete Success

- Observations of transient flame phenomena leading toward steady state flames for at least three different mixture ratios of each of two hydrocarbon fuels and two diluents; and
- Observations of flame extinction at both low and high system Damköhler number for at least three different mixture ratios of each of two hydrocarbon fuels and two diluents; and
- Observations of at least two instances of flame front instabilities each including at least one hydrocarbon fuel; and
- Observations of soot formation.



Other Team Members

- **Sean Yoo**, Princeton doctoral student, graduated with Ph.D. in 2006.
- **Heyang Wang**, Princeton doctoral student, graduated with Ph.D. in 2007.
- **Grunde Jomaas**, Princeton doctoral student, graduated with Ph.D. in 2008.
- **Yuxuan Xin**, 3rd year Princeton doctoral student.
- **Xiaofei Liu**, Rutgers doctoral student, graduated with Ph.D. in 2009.



Test Matrix (1/4)

REQUIRED

Exp Set	Sub Set	Fuel Mixture	Ambient	Flow rate (cc/s)	No. of runs
1	(a)	25%H_2/20%CH_4/55%$\mathbf{N_2}$	21%O_2/79%$\mathbf{N_2}$	(i) 5 (entire duration, 25s) (ii) 10 (entire duration, 25s) (iii) 15 (entire duration, 25s) (iv) 10 (5s)→2 (rest of duration) (v) 5 (5s)→2 (rest of duration)	5
	(b)	20%H_2/25%CH_4/55%$\mathbf{N_2}$		(i) through (v) as shown above	5
	(c)	30%H_2/15%CH_4/55%$\mathbf{N_2}$		(i) through (v) as shown above	5
	(d)	45%H_2/55%$\mathbf{N_2}$		(x) 1 (entire duration, 25s) (y) 2 (entire duration, 25s) (z) 2 (5s)→0.5 (rest of duration)	3
	(e)	45%CH_4/55%$\mathbf{N_2}$		(i) through (v) as shown in (a)	5



Test Matrix (2/4)

REQUIRED Continued…

2	(a)	25%H_2/20%CH_4/55%**He**	21%O_2/79%**He**	(i) 5 (entire duration, 25s) (ii) 10 (entire duration, 25s) (iii) 15 (entire duration, 25s) (iv) 10 (5s)→2 (rest of duration) (v) 5 (5s)→2 (rest of duration)	5
	(b)	20%H_2/25%CH_4/55%**He**		(i) through (v) as shown above	5
	(c)	30%H_2/15%CH_4/55%**He**		(i) through (v) as shown above	5
	(d)	45%H_2/55%**He**		(x) 1 (entire duration, 25s) (y) 2 (entire duration, 25s) (z) 2 (5s)→0.5 (rest of duration)	3
	(e)	45%CH_4/55%**He**		(i) through (v) as shown in (a)	5
					Total: 46



Test Matrix (3/4)

HIGHLY DESIRED

Exp Set	Sub Set	Fuel Mixture	Ambient	Flow rate (cc/s)	No. of runs
3	(a)	20%C_2H_4/80%$\mathbf{N_2}$	21%O_2/79%$\mathbf{N_2}$	(i) 5 (entire duration, 25s) (ii) 10 (entire duration, 25s)	2
	(b)	25%C_2H_4/75%$\mathbf{N_2}$		(i) and (ii) as shown above	2
	(c)	30%C_2H_4/70%$\mathbf{N_2}$		(i) and (ii) as shown above	2
4	(a)	20%C_2H_4/80%**He**	21%O_2/79%**He**	(i) 5 (entire duration, 25s) (ii) 10 (entire duration, 25s)	2
	(b)	25%C_2H_4/75%**He**		(i) and (ii) as shown above	2
	(c)	30%C_2H_4/70%**He**		(i) and (ii) as shown above	2
					Total: 12



Test Matrix (4/4)

DESIRED

Exp Set	Sub Set	Fuel Mixture	Ambient	Flow rate (cc/s)	No. of runs
5	(a)	25%H_2/20%CH_4/55%$\mathbf{CO_2}$	21%O_2/79%$\mathbf{CO_2}$	(i) 5 (entire duration, 25s) (ii) 10 (entire duration, 25s) (iii) 15 (entire duration, 25s) (iv) 10 (5s)→2 (rest of duration) (v) 5 (5s)→2 (rest of duration)	5
	(b)	20%H_2/25%CH_4/55%$\mathbf{CO_2}$		(i) through (v) as shown above	5
	(c)	30%H_2/15%CH_4/55%$\mathbf{CO_2}$		(i) through (v) as shown above	5
	(d)	45%H_2/55%$\mathbf{CO_2}$		(x) 1 (entire duration, 25s) (y) 2 (entire duration, 25s) (z) 2 (5s)→0.5 (rest of duration)	3
	(e)	45%CH_4/55%$\mathbf{CO_2}$		(i) through (v) as shown in (a)	5
6	(a)	20%C_2H_4/80%$\mathbf{CO_2}$		(i) 5 (entire duration, 25s) (ii) 10 (entire duration, 25s)	2
	(b)	25%C_2H_4/75%$\mathbf{CO_2}$		(i) and (ii) as shown above	2
	(c)	30%C_2H_4/70%$\mathbf{CO_2}$		(i) and (ii) as shown above	2
					Total: 29